PennState College of Engineering | KEN AND MARY ALICE LINDQUIST DEPARTMENT OF NUCLEAR ENGINEERING

NUCE 403 – Advanced Nuclear Reactor Design

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Final Project – Groups and Materials

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Materials

The proposed design will be a Pebble Bed Reactor.

Fuel: TRISO, UO2 with 19.75% enrichment.
Moderator: Graphite.
Coolant: FLiBe.

Preliminary assessment consisted on an OpenMC simulation of a pebble bed without any controls. The TRISO particles were homogenized, and the calculated k_{eff} was 1.10182 +/- 0.00030. The objective of the group is to change the reactor geometry to comport guide tubes for control rod insertion. The potential absorber material will be B_4C.